Machine Learning - Graduate Level Assessment 17 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. In Yann LeCun's cake, the cherry on top is
   (a) reinforcement learning
   (b) self-supervised learning
   (c) unsupervised learning
   (d) supervised learning
2. Which one of the following is the main reason for pruning a Decision Tree?
   (a) To save computing time during testing
   (b) To save space for storing the Decision Tree
   (c) To make the training set error smaller
   (d) To avoid overfitting the training set
3. High entropy means that the partitions in classification are
   (a) pure
   (b) not pure
   (c) useful
   (d) useless
4. Statement 1| The softmax function is commonly used in mutliclass logistic regression. Statement 2| The temperature of a nonuniform softmax distribution affects its entropy.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
5. Which one of the following is equal to P(A, B, C) given Boolean random variables A, B and C, and no independence or conditional independence assumptions between any of them?
   (a) P(A | B) * P(B | C) * P(C | A)
   (b) P(C | A, B) * P(A) * P(B)
   (c) P(A, B | C) * P(C)
   (d) P(A | B, C) * P(B | A, C) * P(C | A, B)

True / False (10 marks)

Answer True or False

6. True or False: Residual connections are integral components in both ResNets and Transformers, enhancing model performance and training stability.
7. True or False: The softmax function is indeed commonly used in multiclass logistic regression to model probabilities of multiple classes.
8. True or False: Averaging the output of multiple decision trees helps decrease variance in model predictions.
9. True or False: Grid search runs reasonably slow for multiple linear regression due to the exhaustive evaluation of parameter combinations.
10. True or False: The polynomial degree is a critical factor in determining the trade-off between underfitting and overfitting in polynomial regression.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of deriving the gradient of the objective function $\frac{1}{2}Xw - y_2^2 + \frac{1}{2}\lambda w_2^2$ with respect to w. Include detailed steps and reasoning.
12. Given two Boolean random variables A and B with specific probabilities, analyze how to derive P(A | B) using the provided information. Discuss the implications of these probabilities in a machine learning context.

End of Paper